Program : **Diploma in Instrumentation Engineering**	
Course Code : **5081**	Course Title: **Industrial Instrumentation**
Semester : **5**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L: 3 T: 1 P: 0)**	Periods per semester: **60**

Course Objectives:

- To insight the measurement techniques of Viscosity, density, Specific gravity, and pH measurements.
- To explain different methods for measurement of physical parameters like humidity, moisture and radiation measurements.
- To explain different methods for measurement of physical parameters like force, torque, speed and velocity measurements.
- To insight the measurement techniques of acceleration, vibration, thickness, and miscellaneous measurements.

Course Prerequisites:

Topic	Course code	Course name	Semester
Concept of transducers		Sensors and transducers	3

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe the construction, working principle and applications of different types of Viscosity, density, Specific gravity, and pH measuring instruments.	15	Understanding
CO2	Explain the construction, working principle and applications of humidity, moisture and radiation measuring instruments.	14	Understanding
CO3	Apply different types of transducers for measurement of physical parameters like force, torque, speed and velocity measurements	14	Applying

CO4	Compare and apply different types of transducers for acceleration, vibration, thickness, and miscellaneous measurements	15	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Describe the construction, working principle and applications of different types of Viscosity, density, Specific gravity and pH measuring instruments**		
M1.01	Understand the fundamentals of Viscosity, density, Specific gravity, and pH measurement	5	Understanding
M1.02	Explain the construction and working principle of different types of Viscosity, density, Specific gravity, and pH measuring instruments.	5	Understanding
M1.03	Compare and apply different transducers for Viscosity, density, Specific gravity, and pH measurements	5	Understanding

Contents:

Viscosity measurement- Absolute viscosity - Kinematic viscosity - relative viscosity – units- capillary viscometer- Saybolt Viscometer - Redwood Viscometer - Applications of viscometer.

Density and Specific gravity measurement- Definition- hydrometer- density measurement using LVDT- force balance method- differential pressure method.

pH Measurements -Sorensen's scale–electrodes for pH measurements- Hydrogen electrode, glass electrode- calomel electrode, combined pH electrode- Digital pH meter.

CO2	**Explain the construction, working principle and applications of humidity, moisture and radiation measuring instruments.**		
M2.01	Understand the fundamentals of humidity, moisture and radiation measurement	5	Understanding
M2.02	Explain the construction and working principle of different types of humidity, moisture and radiation measuring techniques	5	Understanding
M1.03	Compare and apply different transducers for humidity, moisture and radiation measurement	4	Understanding
	Series Test – I	1	

Contents:

Humidity measurement-Definition - relative humidity –Absolute humidity - Dew Point- Measurement methods –dry and Wet bulb psychrometer- - Sling psychrometer. - Hair hygrometer – resistive hygrometer - Dew cell - electrolytic hygrometers.

Moisture measurement – measurement of moisture in gases and liquids- capacitance hygrometer- impedance hygrometer. Measurement of moisture in solids- infrared absorption or reflection moisture gauges- conductance or resistance moisture gauges.

Radiation measurement- Types of radiations-

Geiger Muller tube- ionization chamber- scintillation counter- applications of radiation detectors.

CO3	**Apply different types of transducers for measurement of physical parameters like force, torque, speed and velocity measurements**		
M3.01	Understand the fundamentals of force, torque, speed and velocity measurement.	5	Applying
M3.02	Explain the construction and working principle of different types of force, torque, speed and velocity measuring techniques.	5	Understanding
M3.03	Compare and apply different transducers for force, torque, and speed and velocity measurement.	4	Applying

Contents:

Force measurement- Hydraulic load cells – pneumatic load cell – strain gauge load cell- electric force transducer.

Torque measurement- rotating torque sensor- stationary torque sensor- proximity torque sensor.

Speed and velocity measurement – revolution counter- magnetic drag or eddy current tacho meter- AC tachometer generator- DC tacho meter generator- - Stroboscope – translational velocity transducer.

CO4	**Compare and apply different types of transducers for acceleration, vibration, thickness, and miscellaneous measurements.**		
M4.01	Understand the fundamentals of acceleration, vibration, thickness and miscellaneous measurements.	5	Understanding
M4.02	Explain the construction and working principle of different types of acceleration, vibration, thickness, and miscellaneous measuring techniques.	5	Understanding
M4.03	Compare and apply different transducers for acceleration, vibration, thickness, and miscellaneous measurements.	5	Applying
	Series Test – 2	1	

Contents:

Acceleration and vibration measurement- Seismic accelerometer- LVDT accelerometer - Piezoelectric accelerometer – eddy current proximity sensor as vibration pick up- mechanical vibration sensors.

Thickness measurement- inductive methods- eddy current transducer method- capacitive method- ultrasonic method

Miscellaneous measurements using Advanced sensors- Smoke& gas detector, leak detectors, flame detectors, Nanosensors, MEMS.

Text / Reference:

T/R	Book Title /Author
R1	A.K. Sawhney, A course in Mechanical measurements and Instrumentation, Dhanpat Rai and Sons.
R2	S.K Singh, Industrial Instrumentation and Control, TMH.
R3	R.K Jain, Mechanical and Industrial measurements, Khanna Publishers.
R4	Krishnaswami, Industrial instrumentation, New Age Publications.

Online resources:

SL.No	Website Link
1	www.reference.com › Science › Physics › Motion & Mechanics
2	https://study.com/.../what-is-acceleration-definition-and-formula.html